

# Complexity score and known-hazard annotation track different pipeline outcomes in a human pangenome graph

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## Introduction

A pangenome graph represents many haplotypes at once, rather than forcing every sample onto one linear reference. The PanGenome Graph Builder (PGGB) builds such graphs without designating any input as privileged, aligning all haplotypes to one another and constructing the graph directly from that all-to-all alignment (Garrison et al., 2024), and it is one of the methods the Human Pangenome Reference Consortium (HPRC) uses to build its population-scale pangenome graph from long-read haplotype assemblies (Liao et al., 2023; Lucas et al., 2026). *vg giraffe* was introduced as a short-read mapper for graphs of this kind, seeding against a curated set of haplotype paths rather than the whole graph, which is what makes mapping to a graph with hundreds of haplotypes tractable at all (Sirén et al., 2021); long-read mapping was added subsequently (Chang et al., 2025).

Whether that combination scales past a single locus is still an open question. Chang and colleagues, describing the current state of Giraffe, report that they “have not been able to construct indexes for Giraffe” for PGGB graphs at genome scale, and attribute this directly to PGGB’s reference-free topology (Chang et al., 2025). That is a statement about outcome, not mechanism: it does not say what fraction of a genome would be affected, whether difficulty is concentrated in particular kinds of sequence, or whether topological difficulty and mapping inaccuracy are the same failure mode or two different ones.

Existing work on pangenome region difficulty answers adjacent questions. Li defines sample-agnostic “easy” regions for short-read variant calling by k-mer uniqueness across hundreds of assemblies, entirely independent of any graph (Li, 2025). Andreace and colleagues, and Dubois and colleagues, compare graphs built from the same genomes by different construction methods, and both find that the resulting structural disagreement concentrates in tandem repeats and other low-complexity sequence (Andreace et al., 2023; Dubois et al., 2025). None of these studies asks whether a single graph’s own local topology predicts what happens when that graph is indexed and mapped against.

We tested this directly on chromosome 21 of the HPRC release 2 PGGB graph, at a scale small enough to screen exhaustively: every 100 kb window on the chromosome. For each window we computed an explicit topology-based complexity score, and separately annotated known problem sequence classes (segmental duplications, large tandem repeats) from public UCSC tracks. We then asked whether either measure is associated with whether *vg autoindex* and *vg giraffe/vg surject* complete for a region, and whether either is associated with the accuracy of the resulting short-read alignments once they do.

## Methods

### Graph and reference data

We used a chromosome 21 PGGB pangenome graph (parameters p98-k311) built from a pre-release snapshot of HPRC release 2, dated 14 October 2025 and not yet fully quality-controlled or formally published at the time of download, per the HPRC's own pangenome resources repository. The graph contains GRCh38 and CHM13 as reference paths alongside 464 HPRC year-2 haplotypes (466 haplotype/reference paths in total) (Garrison et al., 2024; Lucas et al., 2026); this haplotype count is specific to our snapshot and does not necessarily match later, fully quality-controlled releases of the same resource. HPRC release 2 is, in its completed form, roughly a fivefold expansion, in genome number, over the original draft pangenome (Liao et al., 2023; Lucas et al., 2026). Chromosome 21 was chosen as the smallest human autosome, to keep chromosome-scale extraction and screening tractable on a single workstation. An independent GRCh38 chromosome 21 FASTA (UCSC hg38) was used for N-content screening and as a graph-independent coordinate reference. All graph operations (build, extract, stats, degree, viz) used the odgi toolkit (Guarracino et al., 2022). Full provenance, including checksums and download dates for every input file, is recorded alongside this repository.

### Candidate window screening

Every non-overlapping 100 kb window on chromosome 21 was considered, after excluding the first and last 2% of the chromosome and any window with more than 5% N content in the GRCh38 reference, leaving 367 candidate windows. Each was extracted from the graph with `odgi extract` and summarized with `odgi stats` and `odgi degree`. Windows were dropped from further analysis under two criteria. First, a degenerate-window filter (`edge_count == 0`, or fewer than 400 raw path fragments, or an extracted size more than three times the requested window) removed 22 windows clustered in the pericentromeric and acrocentric part of the chromosome, leaving 345. Second, because the raw path-fragment count used above does not distinguish a haplotype that is finely fragmented from a haplotype that is simply absent, we additionally counted the number of *distinct* haplotypes actually threading each window (parsed from the PanSN `sample#hap` path-name prefix via `odgi paths -L`). Nineteen windows, all within a contiguous stretch immediately flanking the centromere, had fewer than 450 of the 466 possible haplotypes represented and were removed, leaving 326 windows that were carried forward for scoring and sampling.

### Complexity score

For each of the 326 valid windows we computed four graph-topology statistics from `odgi stats/odgi degree`: node density (nodes per kilobase of reference sequence), the edge-to-node ratio, sequence inflation (total graph sequence relative to the reference interval), and maximum node degree. Each statistic was log10-transformed and z-scored across the 326-window population, and the four z-scores were averaged into a single, unitless local complexity score. The score is a purely topological summary: it carries no information about sequence identity or content, which is what distinguishes it from the hazard annotation described next.

### Known-hazard annotation

Complexity score alone cannot see sequence similarity to other parts of the genome, since it is computed from one local subgraph. To capture that separately, every valid window was checked against two UCSC tracks for chromosome 21, `genomicSuperDups` (Bailey et al., 2001) and `simpleRepeat` (Tandem Repeats Finder, (Benson, 1999)), both retrieved once for the whole chromosome and intersected locally rather than queried per window (Kent et al., 2002). A binary hazard flag was set for a window if it overlapped a segmental duplication at

90% identity or higher, or a tandem-repeat array spanning 1.4 kb or more; these thresholds were chosen because they correctly flagged every problem window identified during preliminary, hand-picked testing of this pipeline: two segmental duplications (one inter-chromosomal, one intra-chromosomal) and two large tandem-repeat or NUMT-containing windows. Under this definition, 77 of the 326 windows (23.6%) carry a hazard flag. Crossing the hazard flag with a complexity tier (above or below the population median complexity score) gives a 2x2 design with four strata. Two of the four calibration windows were subsequently drawn into the 50-region evaluation sample described below by the rank-sampling procedure; this is relevant to how the flag's apparent performance in that sample should be read (see Discussion).

## Region sampling and pipeline

Fifty windows were rank-sampled within each of the four complexity-tier-by-hazard-status strata, 25 per tier and 25 per hazard status by construction, so that rare combinations such as a low-complexity window carrying a known hazard would not be missed by chance. Each of the 50 regions was carried through an identical pipeline. Sequence graphs were extracted with `odgi extract` at the default subpath-merging distance; a minority of extractions that expanded past three times the requested window size, a known behaviour of the default merge distance next to large collapsed repeats, were re-extracted with merging disabled. Giraffe indexes were built with `vg autoindex --workflow sr-giraffe` (Sirén et al., 2021). For each region we simulated 30,000 error-free paired-end 150 bp reads (fragment length 350 bp  $\pm$  35 bp) from the GRCh38 interval with `wgsim` (Li, 2011), with a fixed seed for reproducibility. Reads were mapped with `vg giraffe` and surjected onto the linear GRCh38 path with `vg surject` (Sirén et al., 2021); BAM records were inspected with `samtools` (Danecek et al., 2021). A mapped read was scored correct if its absolute position, converted back from the region's local subpath coordinate, fell within 10 bp of either candidate fragment-boundary position implied by the simulated fragment (the read simulator does not record simulated strand, so either boundary is an acceptable match). Every pipeline stage for a region (extraction, indexing, simulation, mapping, surjection, evaluation) was run under a 30-minute wall-clock budget on a single machine (Apple Silicon, 16 cores, 128 GB memory); a region that had not produced a result within that budget was recorded as a pipeline failure. Two genes flagged as biologically notable in this analysis (the ribosomal RNA gene *RNA5-8SN3* and the *MIR6724* microRNA cluster overlapping the single most complex of the 345 scored windows, which was later excluded from the final 326-window pool by the haplotype- representativeness filter) were identified by looking up their coordinates against Ensembl gene annotation (Dyer et al., 2025). All downstream tabulation and the figures in this report were produced with `pandas` (McKinney, 2010) and `matplotlib` (Hunter, 2007). Associations between complexity tier or hazard status and pipeline completion were tested with two-sided Fisher's exact tests; differences in the full accuracy distribution between groups were tested with two-sided Mann-Whitney U tests; both were computed with `SciPy` (Virtanen et al., 2020).

## Results

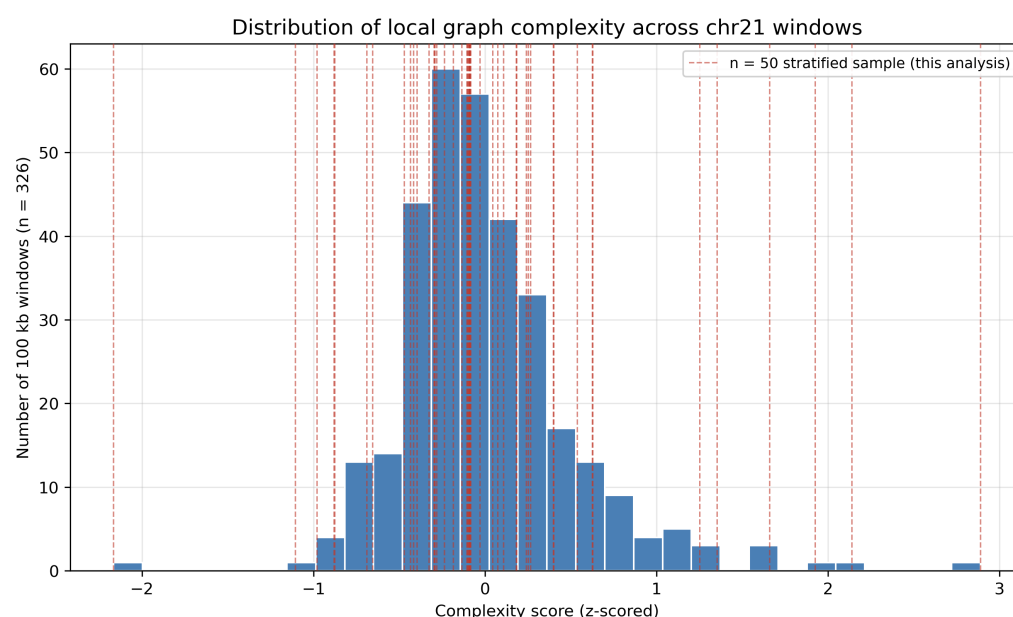
Forty-six of the 50 sampled regions completed extraction, indexing, mapping and surjection within the 30-minute budget; four did not. All four incomplete regions were in the above-median complexity tier: 4 of 25 high-complexity regions did not complete (16%, 95% exact binomial CI 5-36%), compared with 0 of 25 low-complexity regions (0%, CI 0-14%). By hazard status, 3 of 25 hazard-flagged regions did not complete (12%, CI 3-31%) and 1 of 25 regions with no hazard flag did not complete (4%, CI 0-20%). Neither association with completion reached significance in a two-sided Fisher's exact test at this sample size (complexity tier,  $p = 0.11$ ; hazard status,  $p = 0.61$ ).

Among the 46 regions that completed, median accuracy was 99.997% in the low-complexity tier and 99.987% in the high-complexity tier; minimum accuracy in the same two tiers was 98.7% and 99.3%. A two-sided Mann-Whitney U test on the full accuracy distributions found

this difference significant ( $p = 0.04$ ). By hazard status, mean accuracy was 99.92% among regions with no hazard flag and 99.84% among hazard-flagged regions, and minimum accuracy was 99.53% and 98.69%; the corresponding Mann-Whitney U test on the full distributions was not significant ( $p = 0.60$ ). The single lowest-accuracy region in the sample (98.69%) was also the single lowest-complexity region in the 326-window population (complexity score -2.17). This region carried a hazard flag: `genomicSuperDups` lists 41 segmental-duplication entries overlapping the window at 90-97% identity, matching paralogous sequence on six other chromosomes; it is also one of the four windows used to calibrate the hazard-flag thresholds described in Methods.

Two regions encountered during preliminary testing of this pipeline did not complete under a shorter 10-minute budget and completed successfully once given 30 minutes; both are part of this sample. One is the single highest-complexity region tested (complexity score 2.89), which completed at 99.78% accuracy after approximately 26 minutes of total wall-clock time, of which 36 seconds were spent on index construction (the value reported in the per-region results table). The other is a near-median-complexity region (complexity score -0.09) that completed at 99.90% accuracy; its results-table index-construction time (0.39 seconds) reflects the same single pipeline stage, not the longer total time observed for this region during preliminary testing.

Complexity score across the 326-window population was strongly right-skewed, with 77.6% of windows falling within half a standard deviation of the median (Figure 1). The relationships among complexity score, hazard status, pipeline completion, and mapping accuracy across all 50 sampled regions are summarized in Figure 2.



**Figure 1:** Distribution of local complexity score across the 326 valid 100 kb windows on chromosome 21. Dashed lines mark the complexity score of each of the 50 sampled regions



**Figure 2:** Complexity score, pipeline completion, and mapping accuracy across the 50 sampled regions. (a) Complexity score plotted against mapping accuracy, colored by known-hazard status; the four regions that did not complete the pipeline are plotted at the bottom of the axis rather than omitted. (b) Pipeline failure rate by complexity tier (above versus below the population median score). (c) Mean and minimum mapping accuracy among the 46 completed regions, by hazard status

## Discussion

Complexity score and hazard status point toward different outcomes, but at this sample size the formal statistical support for that pattern is mixed rather than clean. Complexity tier shows a large descriptive difference in completion rate (0% failure below the population median score, 16% above it), yet this difference does not reach significance in a two-sided Fisher's exact test ( $p = 0.11$ ); the same is true for hazard status and completion (4% versus 12%,  $p = 0.61$ ). For accuracy, the pattern one would expect from the segmental-duplication case described in Results, that hazard-flagged regions map less accurately, does not hold up under formal testing either: the full accuracy distributions by hazard status are statistically indistinguishable ( $p = 0.60$ ), even though the single lowest-accuracy region in the sample carries a hazard flag. Complexity tier is the one comparison that reaches nominal significance for accuracy ( $p = 0.04$ ); under the two-predictor framing above, accuracy was expected to track hazard status rather than complexity, so the one significant result falls on the measure that was not expected to carry it. The underlying effect is also small, a 0.01 percentage-point difference in median accuracy between tiers, and would not survive a Bonferroni correction for the four comparisons reported here (corrected threshold  $p < 0.0125$ ). None of the four associations tested is robust enough at  $n = 50$  to support a clean division of labor between the two measures.

What the data support more confidently is a single, well-characterized case rather than a population-level rule. A screen built on complexity score alone would have classified this study's lowest-accuracy region, a segmental duplication with 41 paralogous copies elsewhere in the genome, as its safest, because a complexity score computed from a single local subgraph cannot detect sequence identity elsewhere in the genome. That is precisely the property a hazard annotation is designed to catch, and this case is a genuine illustration of the limitation. It should be weighed cautiously, however, because the same region was also one of the four windows used to calibrate the hazard-flag thresholds in the first place (see Methods): using its outcome to argue that the flag generalizes is partly circular, and an independent test set would be needed to show that a hazard flag chosen this way predicts accuracy in regions it was not tuned on.

"Failed to complete" in this study means no result within a fixed 30-minute, single-machine budget, not proof that a region cannot be indexed at all. Two regions in this dataset that initially did not complete under a shorter 10-minute budget completed successfully once given 30 minutes, and the 30-minute budget used throughout this study was chosen using that same observation, so it is not an independently set threshold. The 16% failure rate reported for the high-complexity tier is accordingly better read as an upper bound under this specific, self-selected compute budget than as a fixed property of the graph. The underlying counts are also small (4, 3, and 1 failure events out of 25 regions, respectively), so the wide confidence

intervals reported in the Results should be read alongside the point estimates: the direction of each effect is clearer than its exact magnitude, which is consistent with the non-significant test results reported above.

Chang and colleagues report that Giraffe indexing has not been made to work at genome scale for PGGB graphs (Chang et al., 2025). We do not resolve that problem, but these results sharpen what it looks like below genome scale. Tractability failure at 100 kb is not confined to the single most extreme window on the chromosome: 4 of the 25 above-median-complexity regions sampled here failed to complete under an ordinary compute budget, though the sample is too small to establish how reliably that rate generalizes. A genome-wide extension of Giraffe indexing will need to treat outright failure as an expected outcome to be measured and reported, not an edge case to be debugged away.

Two practical points follow for anyone screening candidate regions at scale, with the caveat above in mind. Existing, cheaply computed sequence annotation, namely the segmental duplication and tandem-repeat tracks that already exist for the human genome, is worth checking alongside any topology-based complexity score, because complexity ranking alone misclassified this study's worst-accuracy region as its safest; whether that failure mode is common enough to matter at scale is a question this sample is too small and too entangled with the calibration set to answer. And because at least two regions in this study that looked like outright failures under a short timeout turned out simply to be slow, whatever timeout a larger screen adopts should be generous enough that "failed" means something more than "ran out of patience."

Most of the limitations below trace back to a single cause: this work was carried out during one BioHackathon, under a fixed and fairly short window of both time and compute, and that constraint shaped the scope of the study as much as any scientific choice did. All extraction, indexing, and mapping ran on one workstation rather than a cluster or cloud allocation, because no other compute was available during the event. The 30-minute per-region timeout, as noted above, was set once two regions were observed needing close to that long to finish, and it was also generous enough to let the full 50-region pipeline complete inside the time remaining in the hackathon; a more patient budget, or more machines run in parallel, would plausibly change the specific failure counts reported above, as those same two regions already demonstrate. Screening was limited to chromosome 21, chosen as the smallest human autosome specifically so that exhaustive 100 kb screening would finish in the available time, so we did not test whether the failure and hazard rates reported here hold on larger, more repeat-rich chromosomes with a different balance of segmental duplication, tandem repeat, and pericentromeric content. Reads were simulated error-free from the linear reference rather than drawn from a real sequencing run, again because acquiring and processing real short-read data at this scale was not feasible inside the hackathon's time budget; this reports graph and mapper behaviour under idealized input, not real Illumina sequencing performance. And the sample of 50 regions, while stratified to guarantee coverage of rare hazard-by-complexity combinations, has a size set by how many regions could be carried through the full pipeline before time ran out, not by a power calculation: the width of the confidence intervals reported in the Results, and the largely non-significant formal test results reported in this section, are a direct, quantitative consequence of that constraint.

One further limitation is not a matter of time or compute. A truth model built from a single simulated coordinate per read cannot distinguish a genuine mapping error from a read placed correctly at one of several biologically valid locations in a duplicated region, precisely the situation a hazard-flagged window is most likely to create, which would bias our accuracy numbers downward rather than up. Real sequencing reads, additional and larger chromosomes, a longer or resource-scaled timeout, and a repeat-aware definition of mapping correctness would each remove one of the constraints above; none of them calls for a different pipeline, only for more time and compute than a single hackathon provides.

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### Author contributions

Kazumichi Fujiwara: Conceptualization, Investigation, Software, Writing – original draft; Yosuke Kawai: Writing – review & editing.

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## Appendices

Full methodology detail, per-region result tables, and the narrative logs documenting the preliminary hand-picked testing referenced above are available in this repository's `README.md` and `logs/` directory.